Practical 12:

CombinedServer.java

```
import java.io.*;

import java.net.*;

public class CombinedServer {

    public static void main(String[] args) {

        int port = 65430;  // common port for all services

        System.out.println("Starting Combined Server on port " + port);


        try (ServerSocket serverSocket = new ServerSocket(port)) {

            while (true) {

                System.out.println("\nWaiting for client connection...");

                Socket clientSocket = serverSocket.accept();

                System.out.println("Client connected from " + clientSocket.getInetAddress());

                BufferedReader in = new BufferedReader(new
InputStreamReader(clientSocket.getInputStream()));

                PrintWriter out = new PrintWriter(clientSocket.getOutputStream(), true);


                // First, read the command from client (hello/file/calc)

                String command = in.readLine();

                if (command == null) {

                    clientSocket.close();

                    continue;

                }

                switch (command.toLowerCase()) {

                    case "hello":

                        handleHello(in, out);

                        break;

                    case "file":
```

```
                handleFileTransfer(clientSocket);
                break;
            case "calc":
                handleCalculator(in, out);
                break;
            default:
                out.println("Unknown command");
        }
        clientSocket.close();
        System.out.println("Client disconnected.");
    }
  } catch (IOException e) {
    e.printStackTrace();
  }
}
private static void handleHello(BufferedReader in, PrintWriter out) throws IOException {
  String msg = in.readLine();
  System.out.println("Received (hello): " + msg);
  out.println("Hello from server!");
}
private static void handleFileTransfer(Socket clientSocket) throws IOException {
  System.out.println("Receiving file...");
  InputStream in = clientSocket.getInputStream();
  FileOutputStream fos = new FileOutputStream("received_file");

  byte[] buffer = new byte[1024];
  int bytesRead;
  // Note: client must close output stream to end transfer
  while ((bytesRead = in.read(buffer)) != -1) {
```

```
        fos.write(buffer, 0, bytesRead);
    }
    fos.close();
    System.out.println("File received and saved as 'received_file'");
}
private static void handleCalculator(BufferedReader in, PrintWriter out) throws IOException {
    System.out.println("Calculator mode started");

    String expression;
    while ((expression = in.readLine()) != null) {
        System.out.println("Calculating: " + expression);
        try {
            double result = evaluate(expression);
            out.println(result);
        } catch (Exception e) {
            out.println("Error: Invalid expression");
        }
    }
}

// Simple evaluator supporting +, -, *, /
private static double evaluate(String expr) throws Exception {
    expr = expr.replaceAll("\\s", "");
    return parseAddSub(expr);
}
// Parser helpers
private static int index = 0;
private static String expression;
private static double parseAddSub(String expr) throws Exception {
```

```
    expression = expr;
    index = 0;
    double value = parseTerm();
    while (index < expression.length()) {
        char op = expression.charAt(index);
        if (op == '+' || op == '-') {
            index++;
            double nextTerm = parseTerm();
            if (op == '+') value += nextTerm;
            else value -= nextTerm;
        } else {
            break;
        }
    }
    return value;
}
private static double parseTerm() throws Exception {
    double value = parseFactor();
    while (index < expression.length()) {
        char op = expression.charAt(index);
        if (op == '*' || op == '/') {
            index++;
            double nextFactor = parseFactor();
            if (op == '*') value *= nextFactor;
            else value /= nextFactor;
        } else {
            break;
        }
    }
```

```
        return value;
    }
    private static double parseFactor() throws Exception {
        StringBuilder sb = new StringBuilder();
        // Handle numbers, including decimals
        if (index < expression.length() && (expression.charAt(index) == '+' || expression.charAt(index) == '-
')) {
            // handle unary + or -
            sb.append(expression.charAt(index));
            index++;
        }
        while (index < expression.length() &&
                (Character.isDigit(expression.charAt(index)) || expression.charAt(index) == '.')) {
            sb.append(expression.charAt(index));
            index++;
        }
        if (sb.length() == 0) throw new Exception("Expected number at position " + index);
        return Double.parseDouble(sb.toString());
    }
}
```

OUTPUT:

```
PS C:\Users\admin\Desktop\CNS>  & 'C:\Program Files\Java\jdk-25\bin\java.exe' '--enable-preview' '-
XX:+ShowCodeDetailsInExceptionMessages' '-cp'
'C:\Users\admin\AppData\Roaming\Code\User\workspaceStorage\2afde5a43ff0d48c99449cdd0a115ab
a\redhat.java\jdt_ws\CNS_859af3d6\bin' 'CombinedServer'

Starting Combined Server on port 65430


Waiting for client connection...

Client connected from /127.0.0.1

Received (hello): How are you
```

Client disconnected.

Waiting for client connection...

Client connected from /127.0.0.1

Receiving file...

File received and saved as 'received_file'

Client disconnected.

Waiting for client connection...

Client connected from /127.0.0.1

Calculator mode started

Calculating: 4*4+6

CombinedClient.java

Code:

```
import java.io.*;
import java.net.*;
import java.util.Scanner;
public class CombinedClient {
    public static void main(String[] args) {
        String host = "127.0.0.1";
        int port = 65430;
        Scanner scanner = new Scanner(System.in); // Create only once here
        while (true) {
            System.out.println("\nChoose service: hello / file / calc / exit");
            System.out.print("> ");
            String command = scanner.nextLine().trim().toLowerCase();

            if (command.equals("exit")) {
                System.out.println("Exiting client.");
                break;
```

```
        }
        switch (command) {
            case "hello":
                runHelloClient(host, port, scanner);
                break;
            case "file":
                runFileClient(host, port, scanner);
                break;
            case "calc":
                runCalcClient(host, port, scanner);
                break;
            default:
                System.out.println("Unknown command, please try again.");
        }
    }
    scanner.close();
}

private static void runHelloClient(String host, int port, Scanner scanner) {
    try (Socket socket = new Socket(host, port);
         BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
         PrintWriter out = new PrintWriter(socket.getOutputStream(), true)) {
        out.println("hello");
        System.out.print("Enter message to send: ");
        String message = scanner.nextLine();  // Use main scanner
        out.println(message);
        String response = in.readLine();
        System.out.println("Received from server: " + response);
    } catch (IOException e) {
```

```
        e.printStackTrace();
    }
}
private static void runFileClient(String host, int port, Scanner scanner) {
    System.out.print("Enter path to file to send: ");
    String filePath = scanner.nextLine();
    try (Socket socket = new Socket(host, port);
         FileInputStream fis = new FileInputStream(filePath);
         OutputStream out = socket.getOutputStream();
         PrintWriter pw = new PrintWriter(socket.getOutputStream(), true);
         BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()))) {

        pw.println("file"); // Send command first

        byte[] buffer = new byte[1024];
        int bytesRead;
        while ((bytesRead = fis.read(buffer)) != -1) {
            out.write(buffer, 0, bytesRead);
        }
        out.flush();
        socket.shutdownOutput();  // signals end of file

        String response = in.readLine();
        if (response != null) {
            System.out.println("Server response: " + response);
        } else {
            System.out.println("File sent successfully.");
        }
```

```
        } catch (IOException e) {
            e.printStackTrace();
        }
    }

    private static void runCalcClient(String host, int port, Scanner scanner) {
        try (Socket socket = new Socket(host, port);
             BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
             PrintWriter out = new PrintWriter(socket.getOutputStream(), true)) {

            out.println("calc"); // Send command first

            String expr;
            while (true) {
                System.out.print("Enter expression (or 'exit' to quit calc): ");
                expr = scanner.nextLine();  // Use main scanner
                if ("exit".equalsIgnoreCase(expr)) {
                    break;
                }
                out.println(expr);
                String response = in.readLine();
                System.out.println("Result: " + response);
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

OUTPUT:

```
PS C:\Users\admin\Desktop\CNS>  & 'C:\Program Files\Java\jdk-25\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\admin\AppData\Roaming\Code\User\workspaceStorage\2afde5a43ff0d48c99449cdd0a115aba\redhat.java\jdt_ws\CNS_859af3d6\bin' 'CombinedClient'


Choose service: hello / file / calc / exit

> hello

Enter message to send: How are you

Received from server: Hello from server!

Choose service: hello / file / calc / exit

> file

Enter path to file to send: C:\Users\admin\Desktop\Doc1.docx

File sent successfully.

Choose service: hello / file / calc / exit

> calc

Enter expression (or 'exit' to quit calc): 4*4+6

Result: 22.0

Enter expression (or 'exit' to quit calc):
```